What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Treatment of Acute Coronary Syndrome – Agents and Interventions

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Adapted from: Treatment of Acute Coronary Syndrome

Listen to this Brick

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Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Outline the treatment goals for acute coronary syndrome (ACS).
2. Justify the pharmacological interventions used in treating ACS.
3. Describe the [illegible] traindications of percutaneous coronary intervention and thrombolytic agents.

CASE CONNECTION

JH, a 63-year-old male with hyperlipidemia, arrives in the emergency department. “This sounds just like the pain my brother had when he had his heart attack. I think I’m having a heart attack!” he tells the attending physician. His ECG findings are consistent with myocardial infarction. Specifically, JH is having an inferior wall ST-segment elevation myocardial infarction (STEMI). You know “time is myocardium,” so you quickly begin your work.

How should JH’s condition be treated? Consider your answers as you read, and we’ll revisit JH at the end of the brick.

GO TO CONCLUSION

The term acute coronary syndrome (ACS) is an umbrella term that includes diagnosis of Unstable Angina, Non-ST Elevation MI (NSTEMI) and ST-Elevation MI (STEMI). This is among the most dramatic and urgent medical problems you will face and understanding its timely and correct treatment is essential.

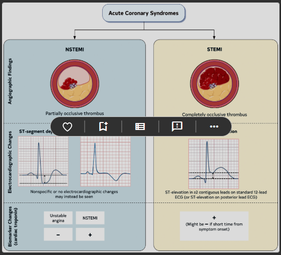
ACS is a grouping of related types of acute ischemic heart disease. But this grouping is not based on their pathology or even test results like ECG or cardiac biomarker results. Rather, all ACS disorders are **urgent** forms of ischemic cardiac disease that usually present with acute chest pain. All undergo **similar initial** treatment in the emergency department. The goal is to restore normal coronary artery flow (and myocardial oxygenation) as soon as possible.

Universal Definition of Myocardial Infarction

We are now on the "Fourth Universal Definition of Myocardial Infarction" as described by the American Heart Association in 2018. This document describes **Myocardial Infarction** as blood levels of **cardiac troponin** (cTn) are increased above the 99th percentile upper reference limit (URL) **in the setting of signs and symptoms of coronary ischemia**. There are actually 5 Types of MI but that is beyond the scope of this discussion.

There are three presentations of ACS:

- **ST-segment elevation MI (STEMI)**, where there is transmural infarction due to complete thrombosis and occlusion of the coronary artery.
- **Non–ST-segment elevation myocardial infarction (NSTEMI)**, where there is less than transmural infarction (nonfull thickness) due to incomplete thrombosis and occlusion of the coronary artery.
- **Unstable angina (UA),** where there is coronary artery thrombosis but no infarction has occurred.


Acute Coronary Syndromes
NSTEMI
STEMI
Angiographic Findings
Partially occlusive thrombus
Completely occlusive thrombus
Electrocardiographic Changes
ST-segment de
ion
Nonspecific or no electrocardiographic changes may instead be seen
ST-elevation in ≥2 contiguous leads on standard 12-lead ECG (or ST-elevation on posterior lead ECG)
Biomarker Changes (cardiac troponin)
Unstable angina
NSTEMI
–
+
+
(Might be – if short time from symptom onset)

Classification of Acute Coronary Syndrome (ACS)

How are these alike and different? All three of these disorders cause occluding thrombi to develop in the coronary arteries, and all can present clinically as a likely MI. This gives rise to the use of common antiplatelet and anticoagulant drugs administered immediately to all patients with ACS. However, not all of them lead to actual infarction of cardiomyocytes, and the ECGs look different on presentation.

What are the goals of patient management? (LO#1)

With a suspected diagnosis of either UA or an acute NSTEMI, the goals of management are to achieve:

- An evaluation of risk
- Initiate antiplatelet and anticoagulant therapy
- Relief from the ischemic pain with nitrates
- β-blocker therapy to prevent ischemia and arrhythmias
- Assessment of h [illegible] abnormalities
- Consideration for invasive (coronary angiography) vs noninvasive strategies (no angiography)

These acute management goals should be followed by therapy that can improve the long-term prognosis for the patient. This will include statins, antiplatelet agents, long term oral anticoagulants, and the use of ACE-I.

The primary goal of STEMI care is to reduce the risk of death and prevent permanent cardiac injury via immediate reperfusion treatment with angioplasty and coronary stenting. Why try to open the artery within two hours after the onset of chest pain symptoms ? Any delay in opening the artery decreases the effectiveness of treatment.

Clinical Significance of Acute Coronary Syndrome

ACS is managed as a medical emergency, usually in emergency departments. Standard treatment protocols are the norm, and the arrival of the patient mobilizes key personnel in the patient care process such as the interventional cardiologist, pharmacist, nurses, etc. The pharmacological agents used in all types of ACS are the same with minor differences in the order of administration.

What are the types of ACS?

What drug classes are used in treating ACS? (LO#2&3)

First, we will look at all the drug classes that are used in ACS. We will discuss the rationale for their use, common ADRs, and indications for the agents.

Morphine

Morphine is an opioid drug traditionally used for pain control during an MI, but studies have shown decreased survival rates and potential interference with absorption of other agents such as P2Y12 inhibitors. Additionally, morphine might mask the patient’s pain without treating its cause, concealing a useful clinical marker of ischemia. *Use it only when nitrates or other analgesics can't relieve severe chest pain.*

Oxygen

Most patients with an MI have no hypoxemia (ie, decreased arterial oxygen) and so do not benefit from oxygen therapy. Research shows that giving oxygen to nonhypoxic MI patients has no benefit and may cause harm. It should only be used when oxygen saturation is less than 90%.

Nitrates

How do Nitrates work? Nitroglycerin produces a molecule (cGMP) that relaxes vascular smooth muscle (veins > arteries). Venodilation, **reduces venous return** to the heart and decreases ventricular preload. The decrease in the contractility and reduction of MVO_2 demand helps in MI. In addition, Nitroglycerin dilates coronary arteries, increasing blood flow directly to the myocardium.

When are they used? Nitrates are administered sublingually in ACS to provide immediate relief from the chest pain and produce the vasodilatory effect. They are used in hemodynamically stable patients (SBP≥90 mm Hg).

What are the potential ADRs? Nitrates can cause hypotension, headaches, and flushing. It is important to note any drug interaction is with phosphodiesterase-5 inhibitors [e.g., Viagra (sildenafil)].

Antiplatelet Medications

Recall that the coronary artery thrombus (clot) formation is the first step that can lead to a myocardial infarction (MI) and hence the primary target is to use agents that prevent clot formation. Aspirin, an antiplatelet agent plays in important role to prevent the clot formation and is important agent in ACS treatment.

How does aspirin work? Aspirin irreversibly inhibits the enzyme cyclooxygenase (COX) and the onset of action is rapid, with the inhibitory effect lasting the life span of the platelet (7-10 days) (Figure 1). All ACS patients should be administered aspirin as soon as possible. In case of STEMI, the aspirin tablet (325 mg) should be chewed or crushed to help increase blood levels as soon as possible. **Note** that the chewing allows fast absorption allowing for platelet inhibition.

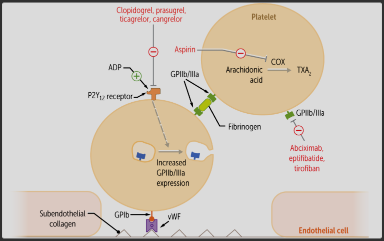


Figure 1

When should aspirin be used? Aspirin remains the gold standard, first-line treatment for a suspected MI. In fact, 9-1-1 operators counsel callers with a suspected MI to take an aspirin, and emergency medical services (EMS) personnel give aspirin during transport. Aspirin (162-325mg orally) should be chewed for a faster onset of action.

What are the potential ADRs? Aspirin also comes with risks, most notably bleeding—especially gastrointestinal bleeding—and each patient's bleeding risk should be evalua

P2Y12 inhibitors: P2Y12 inhibitors include clopidogrel, prasugrel, ticagrelor, and cangrelor.

How do they work? ADP-receptor antagonists block the P2Y12 receptors that normally activate platelets by decreasing intracellular cAMP. (see Figure 1 add fig from antiplatelets). With the inhibition, a platelet plug cannot be formed resulting in **decreased platelet activation.** Clopidogrel and Prasugrel are irreversible inhibitors, while Ticagrelor and Cangrelor are the only ADP-receptor antagonist that inhibit platelets reversibly.

When should they be used? They are initially used in the acute phase of ACS but are often continued for at least 12 months after the acute event.

What are the potential ADRs? Like aspirin, these medications may cause bleeding, and their use should be considered carefully in high-risk patient populations. Ticagrelor has an additional concerning side effect: in about 15% of patients, it causes significant shortness of breath.

When is a patient with MI treated with ADP-receptor antagonists?

Glycoprotein IIb/IIIa Inhibitors. Drugs in this class include abciximab, eptifibatide, and tirofiban.

How do they work? Glycoprotein (GP) IIb/IIIa inhibitors antagonize the GP IIb/IIIa receptor and inhibit the final step in platelet aggregation. This final step is crucial in platelet aggregation because the receptor binds fibrinogen, which leads to linked platelets that form a clot (see Figure 1). These GP IIb/IIIa inhibitors therefore **decrease platelet aggregation.**

When should they be used? Unlike the other drugs in this section, these medicines aren't used routinely in the ER when ACS is suspected. Instead, they are mostly used as an adjunct before or during percutaneous coronary intervention (PCI).

What are the potential ADRs? Adverse effects of GP IIb/IIIa inhibitors include serious bleeding, so they should be used carefully in high-risk patient populations such as those with bleeding disorders, recent gastrointestinal or central nervous system bleeding, or strokes.

What are three common GP IIb/IIIa inhibitors?

Anticoagulants

Anticoagulants are drugs that inhibit the coagulation cascade (Figure 2). This means they inhibit thrombus formation in the setting of ACS, complementing the action of platelet inhibitors. The primary anticoagulant used in the treatment of ACS is heparin. The low molecular weight heparin enoxaparin may also be used, but, given it's longer duration of action, heparin is used more often. These medications have already been covered in depth (Thrombotic disorders foundations and frameworks), so we will only cover them contextually in this brick.

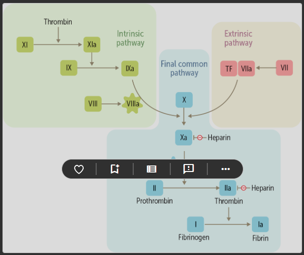


Figure 2

Heparin, like unfractionated heparin and enoxaparin, does not actively work to remove the thrombus; rather, they prevent it from continuing to grow or propagate.

How do they work? Factor X links the intrinsic and extrinsic coagulation pathways; when it converts to Xa, it begins the final common pathway (see Figure 2). Factor IIa (thrombin) is one of the final enzymes in the pathway, responsible for cleaving fibrinogen to fibrin and solidifying the thrombus. Antithrombin inhibits both factor Xa and thrombin. Heparin binds antithrombin and promotes the inactivation of factor Xa and thrombin, preventing clotting.

How are they used in ACS? Heparin can be given intravenously (unfractionated heparin), or as subcutaneous injection in the case of low-molecular-weight heparin (e.g., enoxaparin). The low-molecular-weight drugs have a big advantage: their systemic effect is predictable, so you do not have to steadily monitor the blood prothrombin time (PT) as you do with intravenous heparin infusions. However, as mentioned above, enoxaparin has a longer duration of action and may not be suitable for tenuous patients with high risk factors (e.g., history of bleeding, recent major surgery) for bleeding.

CLINICAL CORRELATION

How long do we give heparin for ACS? Recall that we use antiplatelets and anticoagulants to reduce the chances that the thrombus in the coronary vasculature grows or travels further. This plays a role in anticoagulation therapy because we generally will NOT use anticoagulation for unstable angina, however, for NSTEMI we will give heparin for 48 hours, or until revascularization (i.e., percutaneous coronary intervention in the cath lab) occurs, whichever is first.

What are the potential problems? Heparin increases the risk of bleeding and should be taken with caution by anyone at risk of bleeding, such as a patient with recent surgery or trauma. An absolute contraindication is a history of **heparin-induced thrombocytopenia (HIT)**, which is an immune response causing platelet destruction (a dropping platelet count) and hypercoagulability. HIT occurs more commonly with intravenous heparin.

Other Anticoagulants for Treating ACS. Two alternative anticoagulants can be used to treat ACS, mainly in those allergic to heparin or with HIT.

- Fondaparinux inhibits only factor Xa, like heparins.
- Bivalirudin is a reversible direct thrombin inhibitor. The drug is mostly used when heparin is contraindicated in patients undergoing percutaneous coronary intervention (PCI).

What events in a patient's medical history would make heparin hazardous for patients with ACS?

CLINICAL CORRELATION

What about "MONA"? MONA stands for morphine, oxygen, nitrates, and aspirin. MONA is a traditional mnemonic for treating ACS. But beware! Of those four medications, only aspirin has a proven mortality benefit for ACS. As discussed, morphine, oxygen, and nitrates can all be harmful. So, use evidence-based treatments for ACS instead of MONA.

β-Blockers

These drugs block the β-adrenergic sympathetic receptors and decrease cardiac work and oxygen consumption-useful during an MI.

How do they work? β_1-receptor blockade decreases cardiac workload and decreases oxygen dem s less myocardial ischemia and less chest pain.

The cardioselective, β_1 blocker is preferred. Since cardiac cells only express β_1 receptors, selective agents will prevent side effects such as bronchospasm that can occur with β_2 inhibition in some patients (eg, those with asthma). The agents used are typically metoprolol, atenolol, or esmolol. Metoprolol and atenolol are short-acting, and the dose can be adjusted based on the patient's heart rate and blood pressure. Esmolol is IV and is an ultra-short-acting β-blocker.

When should they be used? β-Blockers are given to all patients with suspected or confirmed MI who can tolerate them in terms of blood pressure and heart rate.

CLINICAL CORRELATION

The mortality benefits of β-blockers have been well documented in STEMI patients. They are recommended as post-STEMI discharge medication.

What are the potential ADRs? β-Blockers can cause sinus bradycardia. In addition, β-blockers may reduce cardiac output by decreasing contractility and heart rate, which might be harmful in patients with heart failure or low cardiac output from an acute MI. Patients with suspected or confirmed cardiogenic shock should not get β-blockers.

β-Blockers that are not cardioselective to β-1 receptors may lead to bronchoconstriction and wheezing in patients with asthma or chronic obstructive pulmonar nificant issue in clinical practice.

Which β-blockers are most effective for patients with MI?

Statins

How do they work? Statins reduce plaque buildup by reducing serum LDL cholesterol levels via inhibition of HMG-CoA reductase. Statins also stabilize atherosclerotic plaques, decrease inflammation, reverse endothelial dysfunction, and decrease thrombogenicity, all of which have been postulated to decrease plaque rupture. These additional benefits are incompletely understood at this time.

When should they be used? If the patient is not on a statin, they are typically started soon after presentation. They should be given as soon as possible and there is data showing benefit if given before stenting in STEMI patients. One of the most powerful be ause mortality and cardiovascular-related mortality immediately following MI. In the acute period, statin's immediate effects are anti-inflammatory and anti-thrombotic, not lipid lowering. Statin therapy provides mortality and morbidity benefits long-term.

What are the potential ADRs? SAMS, hepatoxicity, Avoid in pregnancy

What are the immediate effects of statins in the acute period following MI?

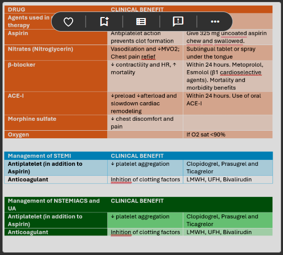
Table 1 reviews the drugs used routinely to lower mortality in patients with ACS. Remember this list!

Table 1

Medication	Mechanisms	Adverse effects
Aspirin	Inhibits cyclooxygenase, decreases platelet aggregation	Bleeding, reaction in patients allergic to aspirin
P2Y12 inhibitors (eg, clopido[illegible] prasugrel, ticagrelor)	[illegible] platelet activation	Bleeding
Fractionated heparin or low-molecular-weight heparins (eg, enoxaparin)	Inhibit factors Xa and IIa, prevent thrombus formation	Bleeding, heparin-induced thrombocytopenia
β-Blockers (eg, carvedilol, metoprolol)	Decrease heart rate and contractility, reduce preload and myocardial oxygen demand	Bradycardia, hypotension
Statins	Lower LDL cholesterol, anti-inflammatory and antithrombotic	Myalgia

Why might morphine be harmful for a patient having an MI?

The clinical benefits for using the agents is summarized in Table-2.



DRUG	CLINICAL BENEFIT	
Agents used in [illegible] therapy		
Aspirin	Antiplatelet action prevents clot formation	Give 325 mg uncoated aspirin chew and swallowed.
Nitrates (Nitroglycerin)	Vasodilation and ↓MVO2; Chest pain refief	Sublingual tablet or spray under the tongue
β-blocker	↓ contractility and HR, ↑ mortality	Within 24 hours. Metoprolol, Esmolol (β1 cardioselective agents). Mortality and morbidity benefits
ACE-I	↓preload ↓afterload and slowdown cardiac remodeling	Within 24 hours. Use of oral ACE-I
Morphine sulfate	↓ chest discomfort and pain	
Oxygen		If O2 sat <90%

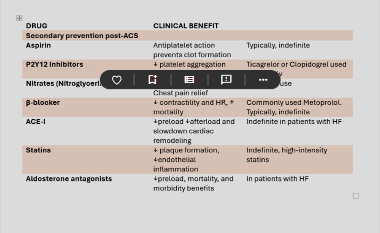
Management of STEMI	CLINICAL BENEFIT	
Antiplatelet (in addition to Aspirin)	↓ platelet aggregation	Clopidogrel, Prasugrel and Ticagrelor
Anticoagulant	Inhition of clotting factors	LMWH, UFH, Bivalirudin

Management of NSTEMIACS and UA	CLINICAL BENEFIT	
Antiplatelet (in addition to Aspirin)	↓ platelet aggregation	Clopidogrel, Prasugrel and Ticagrelor
Anticoagulant	Inhition of clotting factors	LMWH, UFH, Bivalirudin

Table-2: List of Medications used in ACS-STEMI and ACS-NSTEMI

What are the discharge medications used for ACS patients?

These medications listed in Table 3 used post discharge reduce the risk for subsequent MI events and mortality in patients.



DRUG	CLINICAL BENEFIT	
Secondary prevention post-ACS		
Aspirin	Antiplatelet action prevents clot formation	Typically, indefinite
P2Y12 Inhibitors	↓ platelet aggregation	Ticagrelor or Clopidogrel used [illegible]
Nitrates (Nitroglyceri[illegible]	[illegible] Chest pain relief	[illegible] use
β-blocker	↓ contractility and HR, ↑ mortality	Commonly used Metoprolol. Typically, indefinite
ACE-I	↓preload ↓afterload and slowdown cardiac remodeling	Indefinite in patients with HF
Statins	↓ plaque formation, ↓endothelial inflammation	Indefinite, high-intensity statins
Aldosterone antagonists	↓preload, mortality, and morbidity benefits	In patients with HF

Table-3: List of medications post-STEMI/NSTEMI discharge

How Are Interventional Treatments Used to Treat Acute Coronary Syndrome?

The medications we've discussed so far only prevent additional clot formation and reduce myocardial oxygen demand. But how do we actually unclog the thrombosed, occluded coronary artery? Percutaneous coronary intervention (PCI) and chemical thrombolysis (less good) are two typical procedures. The ideal timing for both is—the sooner the better! Thrombolysis and PCI are both considered revascularization therapy and should optimally occur in the first two hours after presentation to the hospital but can be beneficial for up to 12 hours after symptom onset. The effectiveness of revascularization therapy decreases with time.

Percutaneous Coronary Intervention

Percutaneous coronary intervention (PCI), also referred to as percutaneous transluminal coronar nvasive technique used in the management of ACS. PCI uses an intra-arterial wire and ballon to burrow a hole through the coronary artery clot and smash it against the side of the vessel. It is usually done with concurrent stent placement (insertion of a tiny metallic tube to keep the vessel open).

Figure 3 shows an angiogram view of a patient undergoing PCI for an inferior wall MI due to an occluded right coronary artery (RCA). Figure 3A shows a total RCA occlusion (arrow), while Figure 3B demonstrates normal contrast perfusion after PCI.

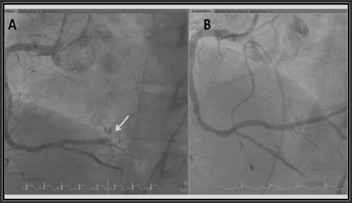


Figure 3

Indications. PCI is the **preferred treatment method** for ST-segment elevation MI. The pri nary artery stenosis of more than 50% in the settings of:

- STEMI
- NSTEMI
- Unstable angina
- Stable angina
- High-risk exercise stress-test findings

The most critical of these is STEMI. In a patient with transmural STEMI, PCI with stent placement **within 90 minutes** of pain onset, and up to 12 hours later, is the treatment of choice. Again, the sooner the better, because complete coronary artery occlusion will allow the area of necrosis to expand progressively over time. With STEMI, time is myocardium. Delay results in more myocardial loss.

The other listed indications are usually less urgent. For NSTEMI (and unstable angina), stent placement is best within 48 hours, but there is much less urgency since the coronary artery is not 100% occluded. The presence of some ongoing perfusion limits the propagation of the subendocardial infarct. Stable angina and high-risk stress tests have no specific time indication for doing PCI.

Evidence of ongoing ischemia in NSTEMI occurs in the minority of patients (about 25%) and includes continuing or recurrent pain, hemodynamic instability, rising troponins, or ventricular dysrhythmias. In these patients, PCI is more urgent.

PCI uses various types of stents. Stents are metallic tubes that have a property of outward radial force that keep the vessel open after deployment much like scaffolding. This allows blood to flow inside the lumen of the stent which improves flow where it was previously limited by a stenosis or clot. The only stents now available h have a drug bonded to a polymer coating attached to the metal of a stent. The drugs limit neointimal proliferation and, hence, the risk of restenosis.

Contraindications. PCI is contraindicated for patients with coagulopathy (bleeding disorder), hypercoagulable states (increased clotting), or diffuse, multiple vessel coronary artery disease.

In addition, when the left main coronary artery (LMCA) lacks collateral flow (as determined by angiography), PCI should not be used. This is because if the LMCA thromboses during the procedure, a large part of the myocardium can infarct. Patients with LMCA lesions instead undergo surgical coronary artery bypass.

Adverse effects. Patients undergoing PCI may experience thrombosis/distal embolization, restenosis, arterial dissection, bleeding and contrast-induced nephropathy (i.e., acute kidney injury from the contrast dye). At catheter insertion sites, hematoma, pseudoaneurysm, or uncommonly limb ischemia may occur.

Why is PCI not done for most patients with occlusion of the left main coronary artery?

Thrombolytic Drugs

Intravenous thrombolytic agents such as **tissue plasminogen activator (tPA) and streptokinase** chemically lyse the obstructing clot in the coronary artery. They are less preferred than PCI but can be used as a backup in STEMI when PCI is not readily available (think of rural hospitals that require a long ambulance ride to get to a PCI-capable facility or a helicopter ambulance that cannot fly due to weather).

How do they work? The body produces a protein called plasminogen that is activated to plasmin which, in turn, breaks down clots. Plasmin does this by cleaving the fibrin bonds that thrombin produced in the coagulation cascade (Figure 4). When tPA is administered, plasminogen is activated to plasmin, which dissolves the clot.

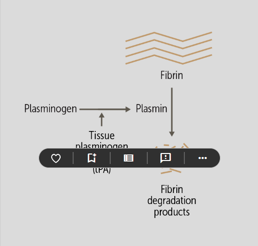


Figure 4

When should they be used? These agents should be used only in STEMI if timely percutaneous intervention (PCI) is unavailable. When used, they should be given within 12 hours of symptom onset—ideally within 30 minutes of first contact with a medical team (e.g., an emergency medical technician or emergency department practitioner), because the effectiveness of these drugs decreases with time.

What are the potential problems? Thrombolytic drugs carry a significant risk of bleeding. Absolute contraindications include:

- Active bleeding or bleeding disorder
- Aortic dissection
- Closed head trauma
- Intracranial hemorrhage
- Recent ischemic stroke
- Severe, uncontrolled hypertension

Thrombolytics are not used for NSTEMI because the occlusion is not complete and studies show no overall improved outcome, therefore the risks (e.g. bleeding) exceed benefits. If ischemia is evolving in NSTEMI and PCI is unavailable, the pati able hospital. There are also many relative contraindications to thrombolytics that go beyond the scope of this discussion.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to JH, what medications should be prescribed?

JH is immediately given aspirin, metoprolol, and heparin and then taken to the heart catheterization lab. The interventional cardiologist places a drug-eluting stent in the right coronary artery, which was 100% occluded. He is now receiving prasugrel. Five days later, you stop to see JH at the time of discharge. He's feeling great, and he thanks everyone for their care of him.

Summary

- Acute Coronary Syndrome (ACS) includes Unstable Angina, Non-ST Elevation MI (NSTEMI), and ST-Elevation MI (STEMI), all presenting with acute chest pain and requiring urgent medical attention.
- The primary goal of ACS treatment is to restore coronary artery flow and myocardial oxygenation quickly, using antiplatelet and anticoagulant drugs.
- The universal definition of Myocardial Infarction involves elevated cardiac troponin levels above the 99th percentile upper reference limit in the context of coronary ischemia symptoms.
- Standard ACS treatment includes antiplatelet medications like aspirin and P2Y12 inhibitors, anticoagulants like heparin, β-blockers to reduce myocardial oxygen demand, and statins for long-term benefits.
- In STEMI cases, immediate reperfusion treatment with angioplasty and coronary stenting is crucial to reduce mortality and prevent permanent cardiac injury; thrombolysis with tPA is an alternative if PCI is unavailable.
- In NSTEMI, immediate reperfusion is less urgent except in the minority of patients with evidence of ongoing ischemia; thrombolysis is not used due to lack of benefit.
- Medications such as morphine and oxygen should be used cautiously in ACS treatment due to potential adverse effects or lack of benefit in nonhypoxic patients.

Review Questions

1. Which of the following medications reversibly blocks the activation of platelets?

A. ○ Aspirin

B. ○ Clopidogrel

C. ○ Heparin

D. ● Ticagrelor

E. ○ Tissue plasminogen activator

Hide Explanation

The correct answer is ticagrelor (D). Ticagrelor is an antiplatelet drug that reversibly binds the adenosine diphosphate receptor and prevents platelet activation. tPA does not affect platelet function. Aspirin (A) and clopidogrel (B) are also antiplatelet medications, but these irreversibly bind COX-2 and ADP receptors, respectively. Heparin (C) is an anticoagulant and does not affect platelet function. Tissue plasminogen activator (E) results in plasmin activation to dissolve clots.

2. A 76-year-old male presents to the emergency department with crushing chest pain. His ECG shows lateral ST-segment elevations. He is given nitroglycerin as part of the treatment regimen, and the chest pain improves. Which of the following best describes the primary mechanism of action of nitroglycerin in reducing his chest pain?

A. ○ Decreases afterload

B. ○ Decreases cardiac contractility

C. ○ Decreases heart rate

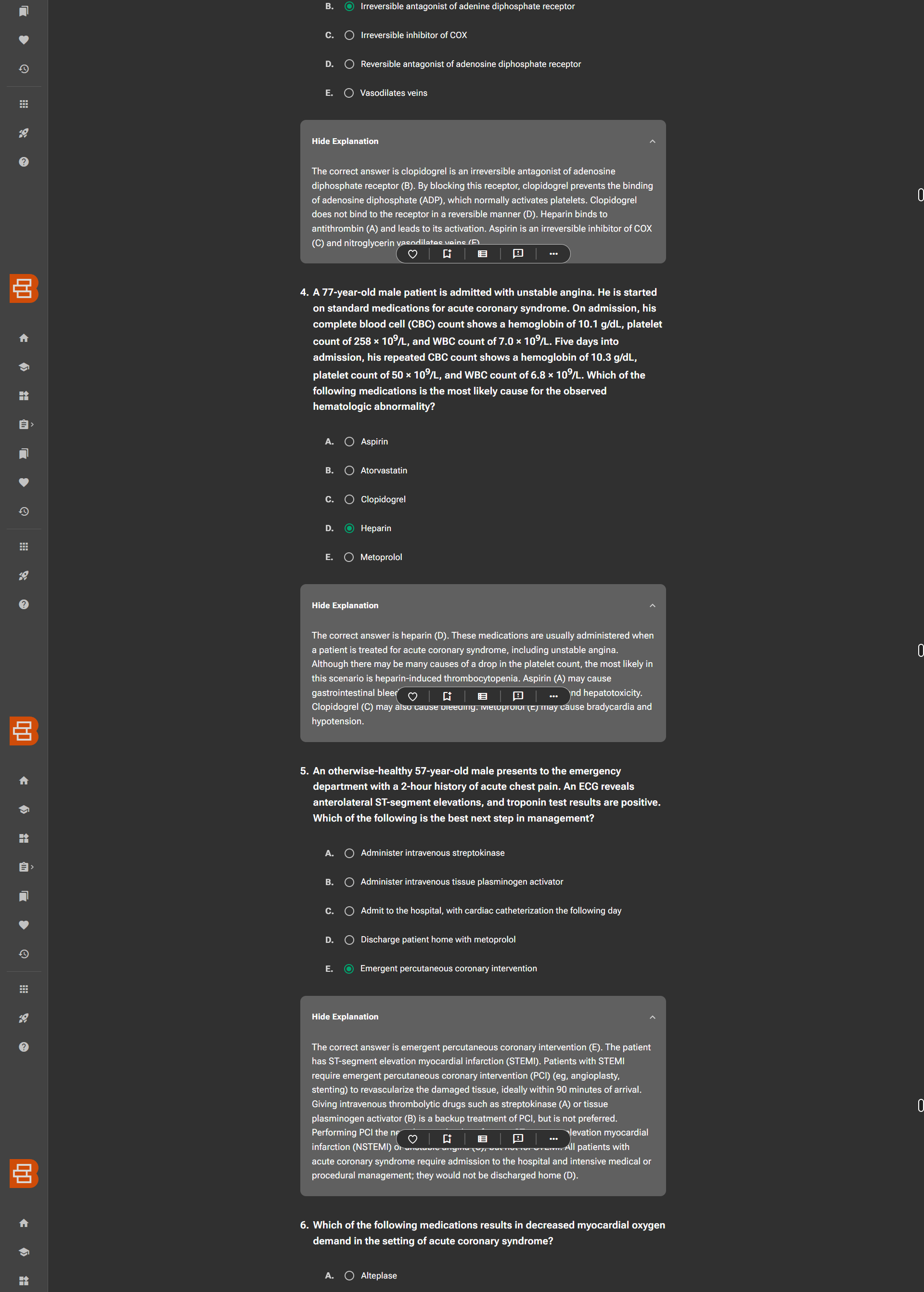
D. ● Vasodilates veins

Hide Explanation

The correct answer is nitroglycerin works by vasodilating veins, also known as "venodilation" (D). Vasodilation of the veins decreases venous return to the heart, thereby requiring it to contract with less contractile force, decreasing myocardial oxygen demand. Altho[illegible]ase in afterload (A), this does not describe the main mechanism of action of the drug. Nitroglycerin also does not directly affect cardiac contractility (B) or directly alter heart rate (C).

3. A 75-year-old male presents to the emergency department and is diagnosed with unstable angina. The patient is given clopidogrel and intravenous heparin. Which of the following statements best describes the mechanism of action of clopidogrel?

A. ○ Binds to antithrombin



B. ● Irreversible antagonist of adenine diphosphate receptor

C. ○ Irreversible inhibitor of COX

D. ○ Reversible antagonist of adenosine diphosphate receptor

E. ○ Vasodilates veins

The correct answer is clopidogrel is an irreversible antagonist of adenosine diphosphate receptor (B). By blocking this receptor, clopidogrel prevents the binding of adenosine diphosphate (ADP), which normally activates platelets. Clopidogrel does not bind to the receptor in a reversible manner (D). Heparin binds to antithrombin (A) and leads to its activation. Aspirin is an irreversible inhibitor of COX (C) and nitroglycerin vasodilates veins (E).

4. A 77-year-old male patient is admitted with unstable angina. He is started on standard medications for acute coronary syndrome. On admission, his complete blood cell (CBC) count shows a hemoglobin of 10.1 g/dL, platelet count of 258 × 10^9/L, and WBC count of 7.0 × 10^9/L. Five days into admission, his repeated CBC count shows a hemoglobin of 10.3 g/dL, platelet count of 50 × 10^9/L, and WBC count of 6.8 × 10^9/L. Which of the following medications is the most likely cause for the observed hematologic abnormality?

A. ○ Aspirin

B. ○ Atorvastatin

C. ○ Clopidogrel

D. ● Heparin

E. ○ Metoprolol

The correct answer is heparin (D). These medications are usually administered when a patient is treated for acute coronary syndrome, including unstable angina. Although there may be many causes of a drop in the platelet count, the most likely in this scenario is heparin-induced thrombocytopenia. Aspirin (A) may cause gastrointestinal blee[illegible]nd hepatotoxicity. Clopidogrel (C) may also cause bleeding. Metoprolol (E) may cause bradycardia and hypotension.

5. An otherwise-healthy 57-year-old male presents to the emergency department with a 2-hour history of acute chest pain. An ECG reveals anterolateral ST-segment elevations, and troponin test results are positive. Which of the following is the best next step in management?

A. ○ Administer intravenous streptokinase

B. ○ Administer intravenous tissue plasminogen activator

C. ○ Admit to the hospital, with cardiac catheterization the following day

D. ○ Discharge patient home with metoprolol

E. ● Emergent percutaneous coronary intervention

The correct answer is emergent percutaneous coronary intervention (E). The patient has ST-segment elevation myocardial infarction (STEMI). Patients with STEMI require emergent percutaneous coronary intervention (PCI) (eg, angioplasty, stenting) to revascularize the damaged tissue, ideally within 90 minutes of arrival. Giving intravenous thrombolytic drugs such as streptokinase (A) or tissue plasminogen activator (B) is a backup treatment of PCI, but is not preferred. Performing PCI the ne[illegible]elevation myocardial infarction (NSTEMI) or unstable angina (C), but not for STEMI. All patients with acute coronary syndrome require admission to the hospital and intensive medical or procedural management; they would not be discharged home (D).

6. Which of the following medications results in decreased myocardial oxygen demand in the setting of acute coronary syndrome?

A. ○ Alteplase

B. ● Metoprolol

C. ○ Heparin

D. ○ Prasugrel

E. ○ Rosuvastatin

Hide Explanation

The correct answer is Metoprolol (B). Metoprolol is a β-blocker, so it antagonizes the β1-adrenergic receptor on myocytes, thereby reducing heart rate and contractility. Reducing these two parameters lowers myocardial oxygen demand. Alteplase (A) is a tissue plasminogen activator that increases thrombus dissolution. Heparin (C) is an anticoagulant that blocks thrombin and factor Xa. Prasugrel (D) is an antiplatelet that antagonizes the adenosine diphosphate receptor and prevents platelet activation, and rosuvastatin (E) is a statin that reduces LDL cholesterol levels.

7. Which is correct regarding the treatment goal for a UA/STEMI patient?

A. ○ Providing acetaminophen for pain relief

B. ● Providing nitrates for pain relief

C. ○ Stopping anticoagulant therapy

D. ○ Stopping β-blocker therapy

Hide Explanation

The correct answer is choice (B) providing nitrate therapy. The administration of nitrates during an ACS event decreases the myocardial oxygen demand on the heart and improves perfusion. This decreases the chest pain associated with an ACS event. Choice (A) is incorrect, since acetaminophen provides COX inhibition and does not directly provide any benefits for chest pain. Choice (C) is incorrect; the appropriate anticoagulant therapy should be initiated and not stopped. Choice (D) is incorrect; β-blocker therapy should be started and not stopped.

How would you rate this Brick?

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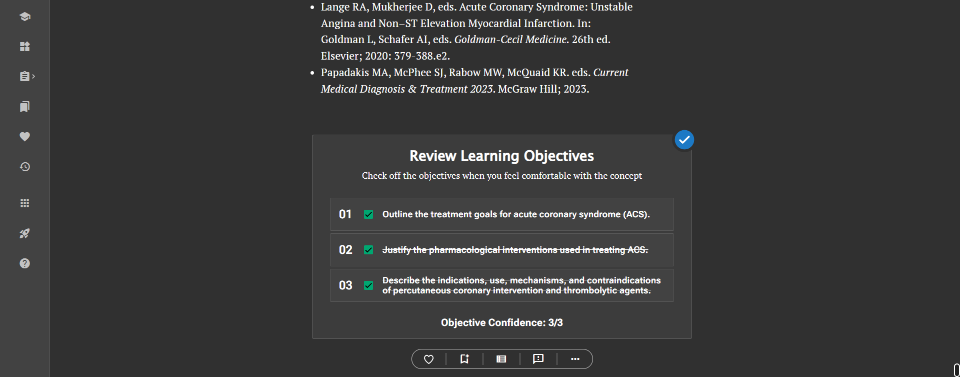
Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- https://www.uptodate.com/contents/overview-of-the-acute-management-of-non-st-elevation-acute-coronary-syndromes?search=acute%20coronary%20syndrome%20treatemnet&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1



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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Outline the treatment goals for acute coronary syndrome (ACS).~~

02 ☑ ~~Justify the pharmacological interventions used in treating ACS.~~

03 ☑ ~~Describe the indications, use, mechanisms, and contraindications of percutaneous coronary intervention and thrombolytic agents.~~

Objective Confidence: 3/3